

Biophysics Homework 1

Mayas Farhat



Question 2: ORFs in E. coli

Part (a): Identification of ORFs via Stop Codon Spacing

The program scans the forward DNA strand in three reading frames (0, +1, +2) to identify stop codons and compute the length L of ORFs — defined as the number of non-stop codons between two consecutive stop codons.

Definitions

Let:

- $\mathcal{S} = \{\text{TAA, TAG, TGA}\}$ be the set of stop codons.
- $p_s = \frac{3}{64}$ be the probability that a randomly selected codon is a stop codon (under uniform nucleotide distribution).
- $q = 1 - p_s = \frac{61}{64}$ be the probability that a codon is *not* a stop codon.

For each reading frame $f \in \{0, 1, 2\}$, the program:

1. Generates the set of codon start positions:

$$\text{codon_starts}_f = \{f, f + 3, f + 6, \dots\} \cap [0, N - 3]$$

where N is the genome length, ensuring only full triplets are considered.

2. Identifies stop codon positions within the frame by scanning codons:

$$\text{stops}_f = \{i \in \mathbb{Z}_{\geq 0} \mid \text{strand}[\text{codon_starts}_f[i] : \text{codon_starts}_f[i] + 3] \in \mathcal{S}\}$$

3. Computes ORF lengths L as distances between consecutive stops:
4. From start of frame to first stop (if any):

$$L_0 = \text{stops}_f[0], \quad \text{if } \text{stops}_f[0] > 0$$

5. Between consecutive stops i and $i + 1$:

$$L_i = (\text{stops}_f[i + 1] - \text{stops}_f[i]) - 1$$

6. From last stop to end of frame (if any):

$$L_{\text{last}} = (\text{total_codons}_f - \text{stops}_f[-1] - 1)$$

The resulting list $\mathcal{L} = \{L_1, L_2, \dots, L_M\}$ contains all ORF lengths across all three reading frames on the forward strand.

Part (b): Comparison with Random Sequence Model

Under the assumption that codons are generated independently and uniformly at random, the distribution of ORF lengths L follows a **geometric distribution**:

$$P(L = \ell) = p_s \cdot q^\ell, \quad \ell = 0, 1, 2, \dots$$

This models the number of non-stop codons before encountering the next stop codon.

Let $\mathcal{L}_{\text{obs}} = \{L_1, \dots, L_M\}$ be the observed ORF lengths from part (a). The empirical frequency distribution is computed as:

$$\hat{P}_{\text{obs}}(\ell) = \frac{\#\{i \mid L_i = \ell\}}{M}, \quad \ell \in \mathbb{Z}_{\geq 0}$$

The theoretical expected count for each ℓ under the geometric model is scaled to match the total number of observed ORFs M :

$$E(\ell) = M \cdot p_s \cdot q^\ell$$

The code plots both $\hat{P}_{\text{obs}}(\ell)$ (as discrete points) and $E(\ell)$ (as a continuous curve) over $\ell = 0, 1, \dots, L_{\text{max}}$, allowing visual comparison between biological data and the random null model.

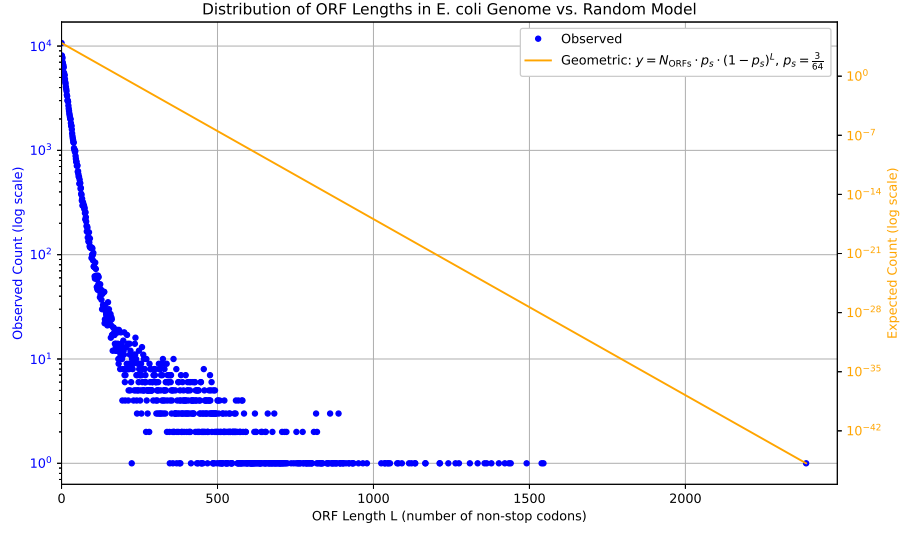


Figure 1: Plot of full ORF distribution

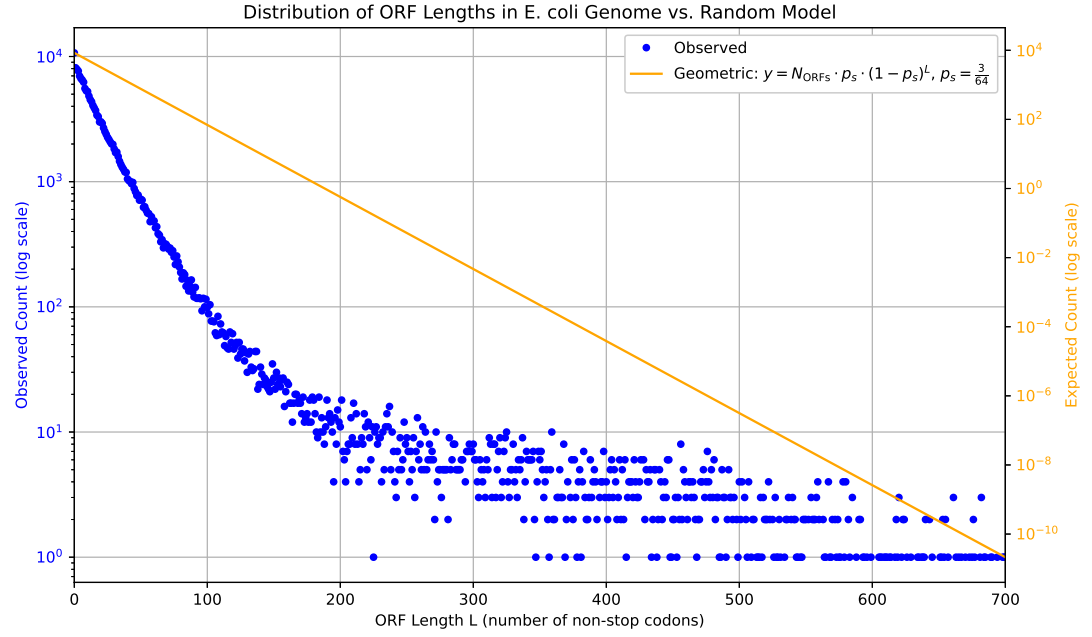


Figure 2: Plot of ORF distribution for $L \leq 700$

Part (c): Estimation of Statistically Significant Cut-off Length L_{cut}

To determine a threshold L_{cut} above which ORFs are unlikely to occur by chance, we compare the **cumulative counts** of observed vs. expected ORFs.

Define:

$$O(L) = \sum_{\ell=L}^{L_{\max}} \#\{i \mid L_i = \ell\}$$

$$E(L) = M \cdot \sum_{\ell=L}^{\infty} p_s \cdot q^\ell = M \cdot q^L$$

The survival function $E(L) = M \cdot q^L$ is derived from the geometric distribution's tail probability:

$$P(L' \geq L) = \sum_{\ell=L}^{\infty} p_s q^\ell = p_s q^L \sum_{k=0}^{\infty} q^k = p_s q^L \cdot \frac{1}{1-q} = p_s q^L \cdot \frac{1}{p_s} = q^L$$

We define statistical significance by choosing a multiplier $k > 1$ (e.g., $k = 10$) and finding the smallest L such that:

$$O(L) > k \cdot E(L)$$

This L is denoted L_{cut} . It represents the minimal ORF length for which the observed frequency exceeds the expected frequency by a factor of k , indicating a statistically significant deviation from randomness — likely due to biological selection for functional genes.

In practice, the code computes $O(L)$ and $E(L)$ for all $L \in \{0, 1, \dots, L_{\max}\}$ and finds the first L satisfying the inequality above.

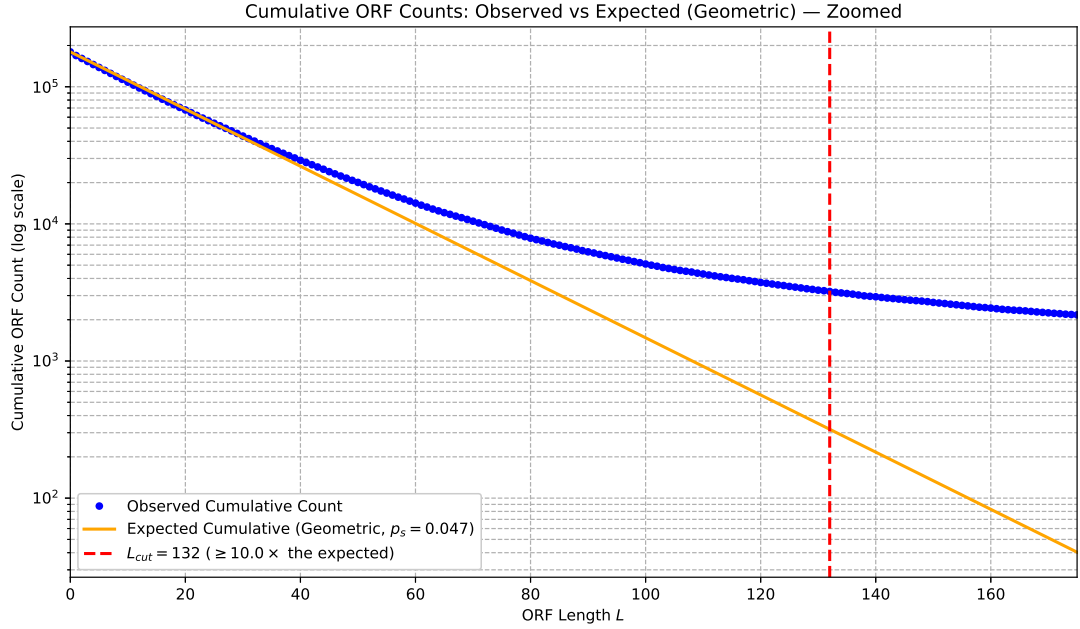


Figure 3: Cumulative distribution of ORF lengths in the *E. coli* genome compared to the expected geometric distribution under a random sequence model. Observed cumulative counts (blue dots) are plotted against the theoretical expectation $N_{ORF} \cdot q^L$ (orange line), where N_{ORF} is the total number of observed ORFs, $q = 1 - p_s$, and $p_s = 3/64$. The vertical red dashed line marks the statistically significant cut-off length $L_{cut} = 132$, above which the observed number of ORFs exceeds $10\times$ the expected count — indicating biological selection for long coding regions.



Question 4 : Correlations in the *E. coli* genome

Part (a): Nucleotide Frequencies

Let the genome sequence be denoted by $S = (s_1, s_2, \dots, s_N)$, where $s_i \in \{A, G, T, C\}$ and N is the genome length. The empirical frequency of nucleotide $X \in \{A, G, T, C\}$ is computed as:

$$p_X = \frac{1}{N} \sum_{i=1}^N \mathbf{1}_{\{s_i=X\}},$$

where $\mathbf{1}_{\{\cdot\}}$ is the indicator function. The code implements this by counting occurrences of each base using `Counter` and normalizing by N . This yields the marginal probability distribution $\{p_A, p_G, p_T, p_C\}$.

For the *E. coli* genome, the computed frequencies are:

A	G	T	C
0.2462	0.2537	0.2459	0.2542

Part (b): Joint Probabilities and Correlation Matrix for Adjacent Bases ($n = 1$)

For neighboring nucleotides separated by distance $n = 1$, the code counts all ordered pairs (s_i, s_{i+1}) for $i = 1, \dots, N - 1$. The joint empirical probability of observing base X followed by base Y is:

$$p_{XY}^{(1)} = \frac{1}{N-1} \sum_{i=1}^{N-1} \mathbf{1}_{\{s_i=X, s_{i+1}=Y\}}.$$

From this, the correlation matrix is defined here as:

$$c_{XY}^{(1)} = \frac{p_{XY}^{(1)}}{p_X p_Y},$$

which quantifies the deviation from statistical independence (where $c_{XY}^{(1)} = 1$). The code constructs a 4×4 matrix with rows and columns indexed by $\{A, G, T, C\}$, computing $c_{XY}^{(1)}$ for all 16 combinations. The resulting correlation matrix for $n = 1$ is:

$$c_{XY}^{(1)} = \begin{matrix} & \begin{matrix} A & G & T & C \end{matrix} \\ \begin{matrix} A \\ G \\ T \\ C \end{matrix} & \begin{bmatrix} 1.2014 & 0.8211 & 1.1030 & 0.8838 \\ 0.9224 & 0.9048 & 0.8831 & 1.2831 \\ 0.7545 & 1.1134 & 1.2099 & 0.9215 \\ 1.1198 & 1.1585 & 0.8138 & 0.9060 \end{bmatrix} \end{matrix}$$

Values greater than 1 indicate over-representation relative to independence (e.g., C followed by G has $c_{CG} = 1.1585$), while values less than 1 indicate under-representation relative to independence (e.g., T followed by A has $c_{TA} = 0.7545$).

Part (c): Correlation Decay with Separation n

The analysis is generalized to arbitrary separation $n \geq 1$. For each n , the code computes:

$$p_{XY}^{(n)} = \frac{1}{N-n} \sum_{i=1}^{N-n} \mathbf{1}_{\{s_i=X, s_{i+n}=Y\}},$$

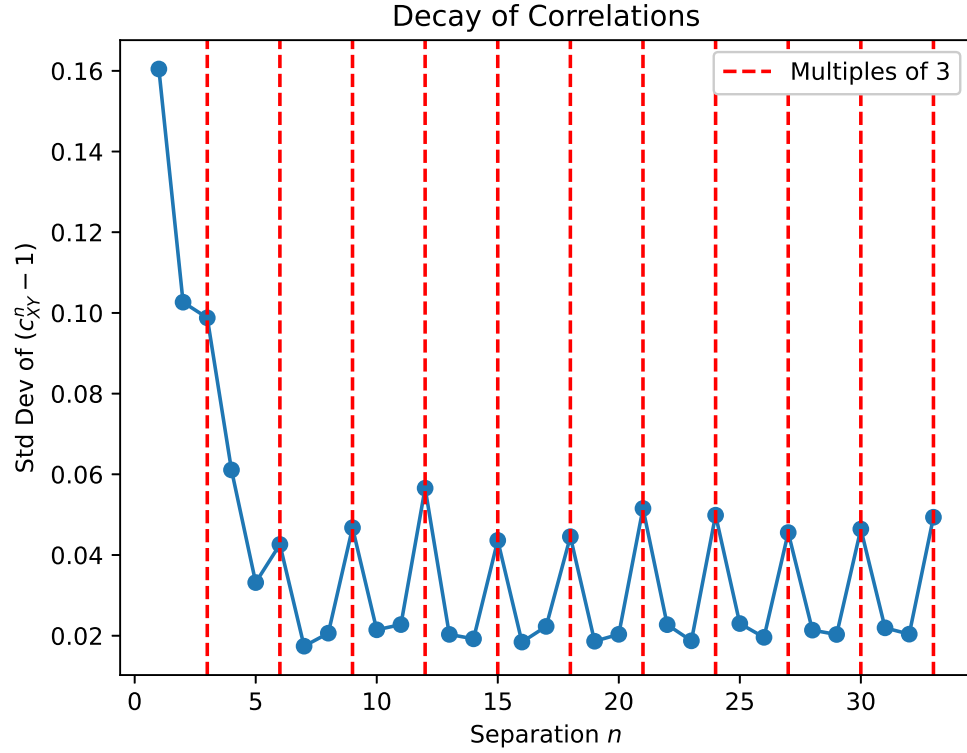
and the corresponding correlation matrix:

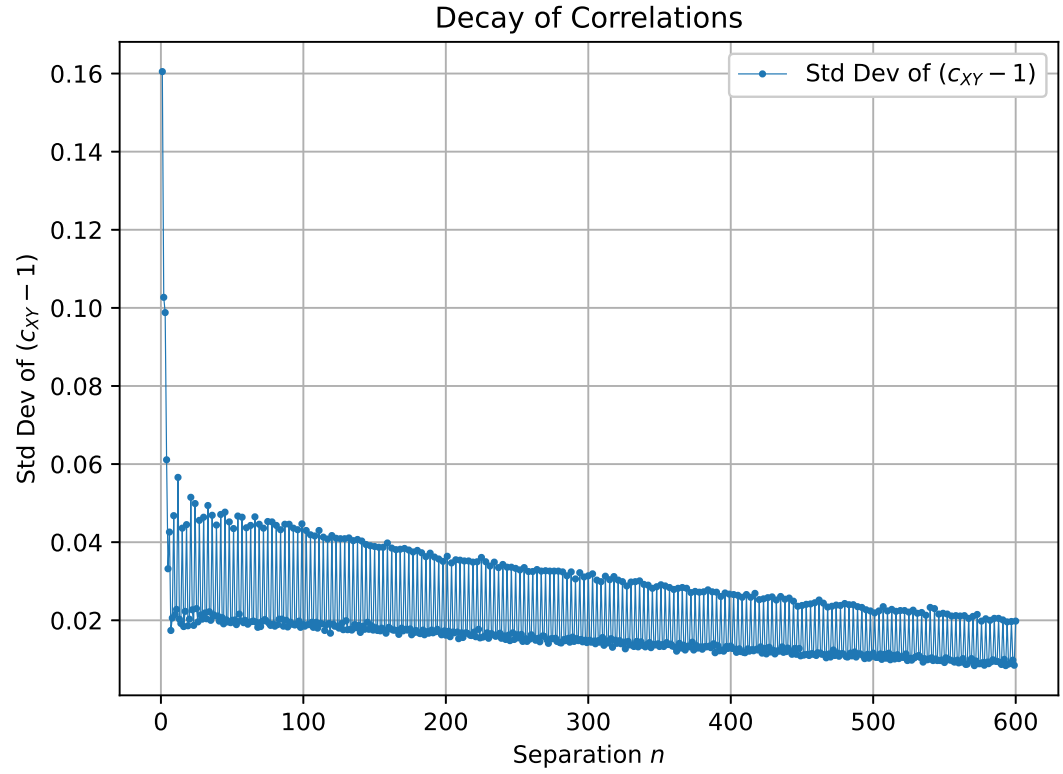
$$c_{XY}^{(n)} = \frac{p_{XY}^{(n)}}{p_X p_Y}.$$

To quantify the overall strength of correlations at distance n , the code computes the standard deviation of the deviation from independence:

$$\sigma_n = \sqrt{\frac{1}{16} \sum_{X,Y \in \{A,G,T,C\}} \left(c_{XY}^{(n)} - 1 \right)^2}.$$

This scalar σ_n measures how much the observed correlations differ from the null hypothesis of independent nucleotides. The code evaluates σ_n for $n = 1$ to $n = N_{\max}$ (e.g., 600) and plots σ_n versus n , thereby characterizing the decay of correlations with genomic distance. The inclusion of vertical lines at multiples of 3 allows inspection of potential codon-periodic structure.





The periodicity seen at $n = 3k$, for integer k , in the E. coli genome comes from the way its genetic code is organized into triplet codons, where each set of three nucleotides codes for a single amino acid. Because the genome is read in these non-overlapping groups of three, a repeating pattern naturally emerges.

We thus have a 3-base periodicity due to codon structure, with a general decay in correlations as n increases.

1 Open Reading Frames

Assumption: A, G, T, C occur with equal probability and independently along a DNA sequence

a) Stop codons = $\{TAA, TAG, TGA\}$ | codon = triplet
= 3 "bases"

Total # of possible codons: $N = 4^3 = 64$ (independence)

Probability to get a particular codon is $\left(\frac{1}{4}\right)^3 = \frac{1}{64}$

Since we have 3 stop codons, the probability of a

stop signal is: $3 \cdot \left(\frac{1}{4}\right)^3 = \frac{3}{64} = \frac{\# \text{ stop codons}}{\# \text{ possible codons}} = P_s$

b) Probability of N non-stop triplets: $\left(1 - \frac{3}{64}\right)^N = \left(\frac{61}{64}\right)^N$

→ $\text{Prob(ORF of length } N) = \left(\frac{61}{64}\right)^N \left(\frac{3}{64}\right) = (1 - P_s)^N P_s$

c) *E. coli* genome has $L = 5 \times 10^6$ bases per strand in a closed loop.

For each reading frame: $M = \frac{L}{3} = \frac{5}{3} \times 10^6$ codons

(frames 1, 2, 3 for top (forward) strand; -1, -2, -3 for bottom (reverse) strand always in 5' → 3' direction).

i.e. M possible starting positions for an ORF (one at each codon) per frame.

We define, for $i = 1, 2, \dots, M$:

$$I_i = \begin{cases} 1, & \text{if an ORF of length 600 starts at position } i, \\ 0, & \text{otherwise} \end{cases}$$

$$\hookrightarrow \mathbb{E} \left[\sum_{i=1}^M I_i \right] = \sum_{i=1}^M \mathbb{E}[I_i] = \sum_{i=1}^M \Pr(I_i = 1) = M(1-p_s)^{600} p_s$$

= expected # of ORFs per frame

Since there are 6 independent frames, each contributing independently under the random base assumption, the total expected # of ORFs across all frames is:

$$6 \cdot \mathbb{E} \left(\sum_{i=1}^M I_i \right) = 6 M (1-p_s)^{600} p_s = 10^7 \left(\frac{61}{64} \right)^{600} \frac{3}{64} \approx 10^{-7}$$

→ On the basis of chance ORFs of length 600 are extremely unlikely. (* Next page is a reminder and can be skipped)

* 1 triplet $\Rightarrow$ instructions for 1 amino acid

For a given DNA sequence (e.g. ATGC GATC...) it matters where we start grouping into triplets i.e. starting at position 1, 2 or 3 gives a different set of codons $\Rightarrow$ 3 possible reading frames per strand. Since we have 2 complementary strands (one "forward" and one "reverse") we have 6 possible reading frames.

Def: A "reading frame" is a way of dividing a nucleotide sequence into codons.

For each gene, only one strand is used as a "template", the other strand is not used for that gene.

Genes can be on either strand, and in practice some are on the "top" strand, others on the "bottom".

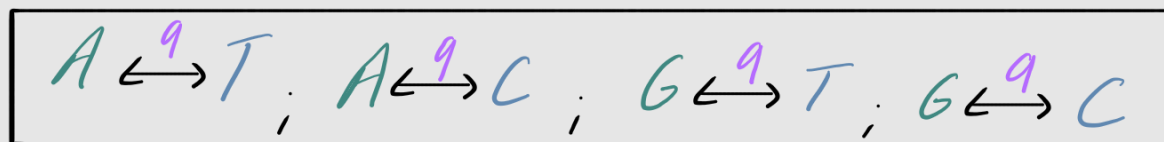
For any particular gene, exactly one frame on exactly one strand will have the proper sequence to code for that gene (start codon + Protein coding sequence + stop codon). The 6 frames are thus the search space for possible protein-coding genes.

By scanning all 6 frames we identify candidate genes in a genome.

3 Point mutations in DNA:

a) Purines: $\{A, G\}$; Pyrimidines: $\{T, C\}$

*) Transversion: Purine $\xleftrightarrow{q}$ Pyrimidine



**) Transitions: Purine $\xleftrightarrow{p}$ Purine : $A \xleftrightarrow{p} G$

Pyrimidine $\xleftrightarrow{p}$ Pyrimidine: $T \xleftrightarrow{p} C$

Prob. that A does not mutate

$$P_A^{i+1} = P_G^i \cdot p + (P_T^i + P_C^i)q + P_A^i(1 - 2q - p)$$

$$P_G^{i+1} = P_A^i \cdot p + (P_T^i + P_C^i)q + P_G^i(1 - 2q - p)$$

$$P_T^{i+1} = P_C^i \cdot p + (P_A^i + P_G^i)q + P_T^i(1 - 2q - p)$$

$$P_C^{i+1} = P_T^i \cdot p + (P_A^i + P_G^i)q + P_C^i(1 - 2q - p)$$

where $p^{i+1} = \begin{pmatrix} P_A^{i+1} \\ P_G^{i+1} \\ P_T^{i+1} \\ P_C^{i+1} \end{pmatrix}$ is the probability vector for generation $i+1$

$$\Leftrightarrow \Pi_1 = \begin{pmatrix} 1-2q-p & p & q & q \\ p & 1-2q-p & q & q \\ q & q & 1-2q-p & p \\ q & q & p & 1-2q-p \end{pmatrix}$$

b) Calculating $\det(\Pi_1 - \lambda I_4) = 0$ by brute force is too long, we follow the hint and use group theory to first find eigenvectors

Suppose $\boxed{P\Pi_1 = \Pi_1 P}$ (P is a symmetry) and $Pv = \lambda v$

$$\rightarrow P(\Pi_1 v) = \Pi_1(Pv) = \lambda(\Pi_1 v) \text{ i.e.}$$

$(\Pi_1 v)$ is also an eigenvector of P with the same eigenvalue $\lambda \rightarrow$ the eigenspace of P is invariant under Π_1

Symmetry transformations for Π_1 :

$$*) P_{AG} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \text{ i.e. } A \leftrightarrow G$$

→ swapping rows 1 and 2 and columns 1 and 2 leaves P_{AG} invariant

$$*) P_{TC} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \text{ i.e. } T \leftrightarrow C$$

$$*) P_{AG} \cdot P_{TC} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \text{ i.e. } A \leftrightarrow G \text{ \& } T \leftrightarrow C$$

*) $\mathbb{I}$ i.e. no swaps.

we guess the orthogonal eigen vectors:

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \text{ (implied simply by normalisation)}$$

$$v_2 = \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \end{pmatrix}, \quad v_4 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ -1 \end{pmatrix}$$

$$\rightarrow \Pi_1 v_1 = v_1 \quad \text{i.e.} \quad \boxed{\lambda_1 = 1}$$

$$\rightarrow \Pi_1 v_2 = \lambda_2 v_2$$

$$\text{Looking at the 1st row} \Rightarrow \boxed{1 - 4q = \lambda_2}$$

$$\rightarrow \Pi_1 v_3 = \lambda_3 v_3 \Rightarrow \boxed{1 - 2(q+p) = \lambda_3}$$

$$\rightarrow \Pi_1 v_4 = \lambda_4 v_4 \Rightarrow \boxed{1 - 2(q+p) = \lambda_4}$$

The eigenvalues are thus:

$$\boxed{\sigma(\Pi_1) = \{1, 1 - 2(q+p), 1 - 4q\}}$$

c) $\Pi_1 = V D V^{-1}$ where the columns of V are the eigenvectors: $V_1 = (v_1 \ v_2 \ v_3 \ v_4)$

To avoid computing V^{-1} we will instead use orthogonal diagonalisation (since Ω_1 is symmetric we know we can do this).

Normalising the eigenvectors gives us an orthonormal eigenbasis: $\mu_i \equiv \frac{v_i}{\|v_i\|}$

$$\rightarrow \mu_1 = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}; \mu_2 = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}; \mu_3 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \end{pmatrix}; \mu_4 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 0 \\ 1 \\ -1 \end{pmatrix}$$

Having μ_i be the columns of V is beneficial

because if $V = \begin{pmatrix} \boxed{\mu_1} & \boxed{\mu_2} & \boxed{\mu_3} & \boxed{\mu_4} \end{pmatrix}$, then

since $\mu_i \cdot \mu_j = \delta_{ij}$ (since we normalised)

We have $(V^T V)_{ij} = \mu_i^T \mu_j = 1$ i.e. $\boxed{V^T V = \mathbb{I}}$

$$\Rightarrow V(V^T V)V^{-1} = V(\mathbb{I})V^{-1} \Rightarrow \boxed{VV^T = \mathbb{I}}$$

$\Leftrightarrow \boxed{V^{-1} = V^T}$ we thus avoid inverting any matrices

$$\Pi_1^t = \underbrace{(VDV^T)(VDV^T) \dots (VDV^T)}_{t \text{ times}} = VD^t V^T$$

$$= (u_1 \ u_2 \ u_3 \ u_4) \begin{pmatrix} \lambda_1^t & 0 & 0 & 0 \\ 0 & \lambda_2^t & 0 & 0 \\ 0 & 0 & \lambda_3^t & 0 \\ 0 & 0 & 0 & \lambda_4^t \end{pmatrix} \begin{pmatrix} u_1^T \\ u_2^T \\ u_3^T \\ u_4^T \end{pmatrix}$$

$$= \sum_{i=1}^4 \lambda_i^t \underbrace{u_i u_i^T}_{\text{outer product} = \text{projection matrix}}$$

$$= \lambda_1^t \frac{1}{4} \left(\begin{array}{cc|cc} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ \hline -1 & -1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{array} \right) + \lambda_2^t \frac{1}{4} \left(\begin{array}{cc|cc} 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 \\ \hline -1 & -1 & 1 & 1 \\ -1 & -1 & 1 & 1 \end{array} \right)$$

$$+ \lambda_3^t \frac{1}{2} \left(\begin{array}{cc|cc} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) + \lambda_4^t \frac{1}{2} \left(\begin{array}{cc|cc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{array} \right)$$

noting $\boxed{\lambda_3 = \mu = \lambda_4}$, we get

$$\mathcal{I}^t = \frac{1}{4} \begin{pmatrix} \lambda_1^t + \lambda_2^t + 2\mu^t & \lambda_1^t + \lambda_2^t - 2\mu^t & \vdots & \lambda_1^t - \lambda_2^t & \lambda_1^t - \lambda_2^t \\ \lambda_1^t + \lambda_2^t - 2\mu^t & \lambda_1^t + \lambda_2^t + 2\mu^t & \vdots & \lambda_1^t - \lambda_2^t & \lambda_1^t - \lambda_2^t \\ \hline \lambda_1^t - \lambda_2^t & \lambda_1^t - \lambda_2^t & \vdots & \lambda_1^t + \lambda_2^t + 2\mu^t & \lambda_1^t + \lambda_2^t - 2\mu^t \\ \lambda_1^t - \lambda_2^t & \lambda_1^t - \lambda_2^t & \vdots & \lambda_1^t + \lambda_2^t - 2\mu^t & \lambda_1^t + \lambda_2^t + 2\mu^t \end{pmatrix}$$

Linear algebra reminder:

7.1.2 Spectral decomposition

The spectral decomposition recasts a matrix in terms of its eigenvalues and eigenvectors. This representation turns out to be enormously useful.

Theorem 3. Let $\mathbf{M}$ be a real symmetric $d \times d$ matrix with eigenvalues $\lambda_1, \dots, \lambda_d$ and corresponding orthonormal eigenvectors $\mathbf{u}_1, \dots, \mathbf{u}_d$. Then:

$$1. \mathbf{M} = \underbrace{\begin{pmatrix} \uparrow & \uparrow & & \uparrow \\ \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_d \\ \downarrow & \downarrow & & \downarrow \end{pmatrix}}_{\text{call this } \mathbf{Q}} \underbrace{\begin{pmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots \\ & & & \lambda_d \end{pmatrix}}_{\mathbf{\Lambda}} \underbrace{\begin{pmatrix} \leftarrow & \mathbf{u}_1 & \rightarrow \\ \leftarrow & \mathbf{u}_2 & \rightarrow \\ & \vdots & \\ \leftarrow & \mathbf{u}_d & \rightarrow \end{pmatrix}}_{\mathbf{Q}^T}.$$

*) $\mathbf{u}_i$ vectors are orthonormal

*) $\mathbf{Q} = \mathbf{V}$ from above

$$2. \mathbf{M} = \sum_{i=1}^d \lambda_i \mathbf{u}_i \mathbf{u}_i^T.$$

Proof. A general proof strategy is to observe that $\mathbf{M}$ represents a linear transformation $\mathbf{x} \mapsto \mathbf{M}\mathbf{x}$ on $\mathbb{R}^d$, and as such, is completely determined by its behavior on any set of d linearly independent vectors. For instance, $\{\mathbf{u}_1, \dots, \mathbf{u}_d\}$ are linearly independent, so any $d \times d$ matrix $\mathbf{N}$ that satisfies $\mathbf{N}\mathbf{u}_i = \mathbf{M}\mathbf{u}_i$ (for all i) is necessarily identical to $\mathbf{M}$.

Let's start by verifying (1). For practice, we'll do this two different ways.

Method One: For any i , we have

$$\mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T\mathbf{u}_i = \mathbf{Q}\mathbf{\Lambda}\mathbf{e}_i = \mathbf{Q}\lambda_i\mathbf{e}_i = \lambda_i\mathbf{Q}\mathbf{e}_i = \lambda_i\mathbf{u}_i = \mathbf{M}\mathbf{u}_i.$$

(*)

Thus $\mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T = \mathbf{M}$.

Method Two: Since the $\mathbf{u}_i$ are orthonormal, we have $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}$. Thus $\mathbf{Q}$ is invertible, with $\mathbf{Q}^{-1} = \mathbf{Q}^T$; whereupon $\mathbf{Q}\mathbf{Q}^T = \mathbf{I}$. For any i ,

$$\mathbf{Q}^T\mathbf{M}\mathbf{Q}\mathbf{e}_i = \mathbf{Q}^T\mathbf{M}\mathbf{u}_i = \mathbf{Q}^T\lambda_i\mathbf{u}_i = \lambda_i\mathbf{Q}^T\mathbf{u}_i = \lambda_i\mathbf{e}_i = \mathbf{\Lambda}\mathbf{e}_i.$$

Thus $\mathbf{\Lambda} = \mathbf{Q}^T\mathbf{M}\mathbf{Q}$, which implies $\mathbf{M} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$.

Now for (2). Again we use the same proof strategy. For any j ,

$$\left(\sum_i \lambda_i \mathbf{u}_i \mathbf{u}_i^T \right) \mathbf{u}_j = \lambda_j \mathbf{u}_j = \mathbf{M}\mathbf{u}_j.$$

Hence $\mathbf{M} = \sum_i \lambda_i \mathbf{u}_i \mathbf{u}_i^T$. □

*) Recall :

$$\begin{pmatrix} \boxed{a_{11}} & \dots & \boxed{a_{1n}} \\ \vdots & & \vdots \\ \boxed{a_{m1}} & \dots & \boxed{a_{mn}} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix} = \begin{pmatrix} a_{11}x_1 + \dots + a_{1n}x_n \\ \vdots \\ a_{m1}x_1 + \dots + a_{mn}x_n \end{pmatrix} = x_1 \vec{A}_1 + \dots + x_m \vec{A}_m$$

$\vec{A}_1 \quad \vec{A}_m$

<https://cseweb.ucsd.edu/~dasgupta/291-unsup/lec7.pdf>

d)*) By definition the "steady state" distribution, P_{ss} , is given by $\Pi_1 P_{ss} = P_{ss}$ i.e the corresponding eigen value is 1. We already showed that the associated eigen-vector for $\lambda=1$ is $\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$ and so $P_{ss} \propto \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$

The transition probability matrix determines how probabilities evolve from one generation to the next: $P(i+1) = P^{i+1} = \Pi_1 P^i = \Pi_1 P(i)$

$$\Rightarrow P(t) = \Pi_1^t P(0) = \left[\sum_{k=1}^4 \lambda_k^t (u_k u_k^T) \right] P(0)$$

since $\lambda_1=1$, $\lambda_{2,3,4} < 1$: $\lim_{t \rightarrow \infty} P(t) = (u_1 u_1^T) P(0)$

$$\lim_{t \rightarrow \infty} P(t) = \frac{1}{4} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} P_A(0) \\ P_B(0) \\ P_T(0) \\ P_C(0) \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{pmatrix} = P_{ss}$$

regardless of $P(0)$ (only constraint is normalisation and non-negative entries) we always tend to P_{ss} when $t \rightarrow \infty$.
 In the **steady-state**, all nucleotides appear with probability $1/4$.

***) Estimating the # of generations:

$$P(t) = \left[\sum_{k=1}^4 \lambda_k^t (u_k u_k^T) \right] P(0) = 1^t \underbrace{u_1 u_1^T P(0)}_{P_{ss}} + \sum_{k=2}^4 \lambda_k^t \underbrace{c_k u_k}_{= q_k}$$

where, for $k=2,3,4$: $\boxed{c_k} = u_k^T P(0) = u_k \cdot P(0) \in \mathbb{R}$ (scalar product)

$\Rightarrow$ Deviation from equilibrium at a given t :

$$P(t) - P_{ss} = \lambda_2^t q_2 + \lambda_3^t q_3 + \lambda_4^t q_4$$

for $\underline{i=2,3,4}$: $\lambda_i^t = e^{t \ln \lambda_i}$, since $\lambda_i < 1$: $\ln \lambda_i < 0$

Defining $\frac{1}{\tau_i} = -\ln(\lambda_i) \Rightarrow \lambda_i^t = e^{-\frac{t}{\tau_i}}$

The approach to steady state is governed by the **slowest-decaying transient mode (SDTM)**, associated with the eigen-value closest to 1, denoted λ_{max} , i.e.

$$\lambda_{\max} = \max(\lambda_2, \lambda_3 = \lambda_4) = \max(1-4q, 1-2p-2q) \quad (\lambda_i > 0)$$

because its contribution $\lambda_{\max}^t$ decreases most gradually.

For $\lambda_i \simeq 1$ (valid when $p \ll 1$ and $q \ll 1$, a reasonable assumption when modelling biological mutation rates) we have:

$$-\ln(\lambda_i) = -\ln(1 - \underbrace{(1-\lambda_i)}_{\simeq 0}) \simeq (1-\lambda_i) \Rightarrow \frac{1}{\tau_i} = (1-\lambda_i)$$

$$\Rightarrow \lambda_i^t \simeq e^{-t(1-\lambda_i)}$$

The relaxation time τ is the time it takes for (SDTM) to decay to $\frac{1}{e} \simeq 0,37$:

$$\lambda_{\max}^{\tau} = e^{-1}$$

Note: for a transient mode decaying as $A(t) = A_0 e^{-t/\tau}$:

$$A(\tau) = 0,37 A_0, A(2\tau) \simeq 0,135 A_0, A(3\tau) = 0,05 A_0 \text{ i.e. } 95\% \text{ decayed.}$$

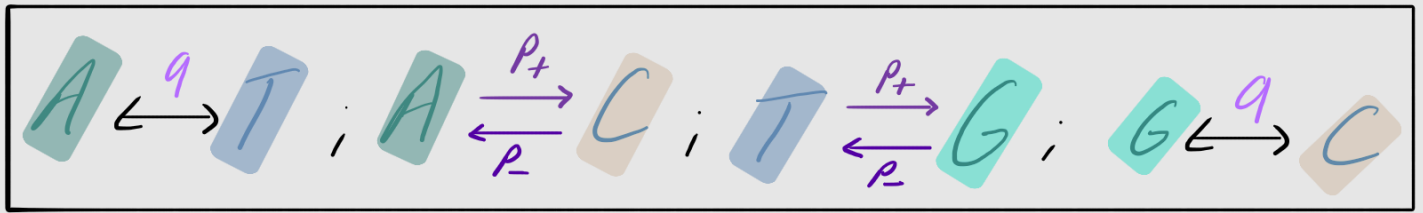
thus when $t \gtrsim \tau$ we consider by convention the system has reached quasi-steady state behavior.

The conventional relaxation time is thus given by

$$\tau(1-\lambda_{\max}) = 1 \Rightarrow \tau = \frac{1}{1-\max(1-4q, 1-2q-2p)}$$

$$r = \begin{cases} \frac{1}{4q}, & \text{if } 4q < 2(q+p) \text{ i.e. } p > q \\ \frac{1}{2(q+p)}, & \text{if } q > p \end{cases}$$

e) With $p_+ > p_-$



$$\begin{pmatrix} p_A(i+1) \\ p_G(i+1) \\ p_T(i+1) \\ p_C(i+1) \end{pmatrix} = \begin{pmatrix} 1-p-q-p_+ & p & q & p_- \\ p & 1-p-q-p_- & p_+ & q \\ q & p_- & 1-p-q-p_+ & p \\ p_+ & q & p & 1-p-q-p_- \end{pmatrix} \begin{pmatrix} p_A(i) \\ p_G(i) \\ p_T(i) \\ p_C(i) \end{pmatrix}$$

For the steady state: $p^* = \mathbb{I} p^*$ where $p^* = \begin{pmatrix} A \\ B \\ C \\ D \end{pmatrix}$

$$pB + qC + p_-D = (p+q+p_+)A \quad (\text{I})$$

$$pA + p_+C + qD = (p+q+p_-)B \quad (\text{II})$$

$$qA + p_-B + pD = (p+q+p_+)C$$

$$p_+A + qB + pC = (p+q+p_-)D$$

We notice that we have a symmetry when swapping $A \leftrightarrow T$ and $G \leftrightarrow C$ we expect that

$$A = P_A^* = P_T^* = C \quad \text{and} \quad B = P_G^* = P_C^* = D$$

(The stationary distribution of an irreducible Markov chain is unique by Perron-Frobenius theorem):

$$\rightarrow (P + P_-)B = (P + P_+)A \quad (I)$$

$$\text{using normalisation: } 2(A+B) = 1 \Rightarrow B = \frac{1}{2} - A$$

$$\hookrightarrow (P + P_-) \left(\frac{1}{2} - A \right) = (P + P_+)A$$

$$\frac{P + P_-}{2} = (2P + P_- + P_+)A$$

$$\rightarrow A = \frac{P + P_-}{2(2P + P_- + P_+)} = P_A^* = P_T^*$$

$$P_G^* = P_C^* = B = \frac{1}{2} - A = \frac{1}{2} \left(1 - \frac{P + P_-}{2P + P_- + P_+} \right) = \frac{P + P_-}{2(2P + P_+ + P_-)}$$

5 Activation/Deactivation reaction:

a)

2 possible states A & B with exchange rates a and b $A \xrightleftharpoons[b]{a} B$

$p(n,t)$:= probability that $N_A = n$ at time t ,

and $N_B = N - n$ since $N = N_A + N_B$ fixed

Different transition rates:

Into n from $n+1$: one of the $(n+1)$ A's turns into B $\rightarrow a(n+1)$

Into n from $n-1$: one of the $(N-(n-1))$ B's turns into A $\rightarrow b(N-n+1)$

Out of n to $n-1$: one of the n A's turns into B $\rightarrow an$

Out of n to $n+1$: one of the $(N-n)$ B's turns into A $\rightarrow b(N-n)$

Thus the Master equation, which describes the rate of change of $p(n,t)$, as the sum of rates of transitions into n from other states minus the rates of transitions out of state n , takes the following form: for $0 < n < N$

$$\frac{dp(n,t)}{dt} = a(n+1)p(n+1,t) + b(N-n+1)p(n-1,t) - [an + b(N-n)]p(n,t)$$

With boundary conditions: $p(-1,t) = 0 = p(N+1,t)$

$$n=0: \quad p(0,t) = a p(1,t) - b N p(0,t)$$

$$n=N: \quad \frac{dp(N,t)}{dt} = b p(N-1,t) - a N p(N,t)$$

b) Initially, all molecules are in state A $p(t=0) = \delta_{m,1} N$.

The molecules are independent, the system is equivalent to N independently evolving binary states i.e. N identical, non-interacting two state Markov processes.

Let $P_A(t)$ denote the probability that a single molecule is in state A at time t ($\Rightarrow P_B(t) = 1 - P_A(t)$). We therefore have the following master equation for this single binary element:

$$\frac{dP_A(t)}{dt} = -aP_A(t) + bP_B(t) \Rightarrow \frac{dP_A(t)}{dt} + (a-b)P_A(t) = b$$

We use the method of integrating factors to solve this 1st order linear ODE i.e. we multiply it by $I = e^{\int (a-b) dt}$.

We recast it using $P_A(t) \equiv y(t)$:

$$\Leftrightarrow e^{(a+b)t} \frac{dy(t)}{dt} + e^{(a+b)t} (a+b)y(t) = b e^{(a+b)t}$$

$$\Leftrightarrow \frac{d}{dt} [y(t) \cdot e^{(a+b)t}] = b e^{(a+b)t}$$

$$\int_0^t \frac{d}{dt'} [y(t') \cdot e^{(a+b)t'}] dt' = \int_0^t b e^{(a+b)t'} dt'$$

$$e^{(a+b)t} y(t) - \underbrace{y(0)}_{=1} = \frac{b}{(a+b)} \left[e^{(a+b)t} - 1 \right]$$

$$e^{(a+b)t} y(t) = \frac{b}{a+b} e^{(a+b)t} + \frac{a}{a+b}$$

$$\Rightarrow P_A(t) = y(t) = \frac{1}{a+b} \left[b + a e^{-(a+b)t} \right]$$

$$\hookrightarrow P_B(t) = 1 - P_A(t) = \frac{a}{a+b} \left[1 - e^{-(a+b)t} \right]$$

Now, the probability that exactly n molecules are in state A (and $N-n$ in state B) at time t follows a binomial distribution, where the "success" probability (being in state A) is $P_A(t)$:

$$\Rightarrow \boxed{p(n,t)} = \binom{N}{n} [P_A(t)]^n [1 - P_A(t)]^{N-n}$$

$$= \binom{N}{n} [P_A(t)]^n [P_B(t)]^{N-n}$$

$$= \boxed{\binom{N}{n} \frac{[b + a e^{-(a+b)t}]^n [a(1 - e^{-(a+b)t})]^{N-n}}{(a+b)^N}}$$